SMITHIAN FOLD THEORY V3

There Is No Nothing

A premise-free operational foundation and an open verification platform for Smithian Fold Theory

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Contents

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> ***Publication record.*** *Maria Smith authorised this status-only correction. It is published open access at 10.5281/zenodo.21761649 through Zenodo's new-version route after 10.5281/zenodo.21760430 in the same concept DOI lineage. The correction changes no scientific claim, evidence classification, mathematical value, source identity or machine record.*
>
> ***Lean verification integration.*** *The current SFT ordered census passed the independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims, 898,902 candidate records and decisions, 11,108 controls, seventeen branches and no reported issue. Lean natively proves the two-class operational root and its unique survivor and constructs proof-bearing acceptance-gate certificates. The remaining claim content is checked as the complete registered artifact graph; that distinction is retained throughout this successor.*

Abstract

This paper states the operational problem of presenting nothing, preserves the premise-free SFT methods constitution and now adds a kernel-checked Lean 4 formalisation of the exact two-class root. Lean proves constructively that `presentedOccurrence` is the unique survivor, with no imported or user-declared axiom reported for the exported root theorem. The compiled verification layer separately accepts all 2,777 current model claims after checking the complete artifact graph.

The paper also specifies a fail-closed admission engine intended to govern the future reconstruction. Every claim must register its dependency chain, provenance class, generated candidate grammar, exact boundary, completeness certificate, decision for every candidate, unique survivor, minimality, named-shape uniqueness, adverse controls and a distinct implementation check. Natural-science claims add preregistration, target-inaccessible prediction, pre-comparison sealing, separate target custody, post-seal measurement, falsification criteria and preservation of unfavourable results. One public command validates repository policy, unit and end-to-end behaviour, 100% of the core engine's executable lines, every registered derivation, controls, independent validators, receipts and the ordered census.

The percentage is a precise software-coverage statement, not a claim that tests eliminate all possible defects.

The platform is a response to documented structural weaknesses in scientific communication and selection: paywall and author-fee inequality, publication bias against null results, sponsor-associated outcome and agenda effects, variable reliability in grant review, and the limited scientific interpretability of black-box prediction. It does not replace domain expertise, peer criticism, measurement or conventional correspondence. It removes credentials, consensus and predictive opacity from the *admission rule* for an SFT law. Code is released under Apache 2.0; papers and documentation are released under CC BY 4.0. Maria Smith retains copyright. The Ernos Labs name is a conformance designation, not a restriction on lawful copying, forking, testing or criticism.

Keywords: Smithian Fold Theory; open science; computational proof; premise-free foundation; exact arithmetic; reproducibility; falsifiability; clean-room replication; scientific software; knowledge tree.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned `PASS` for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

Current status, evidence language and reader map

Status field	Current position
Paper and completion boundary	methodological or comparison scope; no downstream claim ownership transferred. Completion is dated to the current registered census and remains open to lawful versioned extension.
Formal status	The Lean root theorem is kernel checked. Every live model claim has a current admitted receipt and passed the Lean artifact gates. Formal admission does not imply empirical confirmation.
Empirical status	Claim specific. Blind, non-blind, development-observed, holdout, favourable, adverse, null, missing, unavailable, disputed and unresolved distinctions remain exactly as recorded.
Chronology	Claim-specific registration, sealing, custody and observation order controls. Later evidence never becomes an earlier prediction without the registered chronology.
Publication status	Version 0.4.1 is published open access at 10.5281/zenodo.21761649 in the existing Zenodo concept DOI lineage.
Ownership	This paper retains its declared scientific boundary. Lean verification creates no new branch ownership and no application may select a law.
What is not claimed	Permanent closure of science, universal empirical confirmation, conversion of compatibility into confirmation, or superiority to every rival model.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the named formal, computational or empirical test; its kind must be stated.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications; none substitutes for another.
Foundational closure	Completion of the registered foundation only; not field-wide closure.
Field-wide closure	Completion of the named frozen or registered dated field census.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may enter a later version without rewriting receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relations; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. They remain authoritative where prose abbreviates display.

Editorial change control

This successor improves currency, expression, typography and navigation. It does not change scientific meaning, chronology, evidence class, branch ownership, claim status or machine identity. Any conflict between authoritative records must be resolved by explicit supersession or reported to Maria Smith; prose alone cannot manufacture agreement.

Results first: the method, its first proofs and the branch series it enabled

Headline result	Exact result	Meaning
Premise-free operational root	A purported counterexample either presents nothing and supplies no counterexample, or presents an occurrence and is not nothing.	The two-class operational census closes without an axiom, free parameter or imported ontology.
Structural One	Of six complete coverage/addition forms, exactly one retains the admitted occurrence completely without omission or unforced addition.	One is a forced structural self-whole, not a conventional numeral assumed in advance.
Public fail-closed admission method	Registration, complete candidate generation, one decision per candidate, exactly one survivor, closure, adverse controls, independent reconstruction, sealing and-where nature is claimed-post-seal target custody are all mandatory.	A claim cannot enter through reputation, consensus, a successful example, a terminal oracle score or an edited favourable result.
Open scientific authority	Criticism is unrestricted; admission requires the public evidence chain and unchanged-engine receipt. Copyright preserves Maria Smith's authorship while CC BY 4.0 and Apache-2.0 preserve open inspection and reuse.	Ernos Labs is an open-source science movement and a separate revocable standards designation, not a proprietary gate on knowledge.
Current paper series	The inaugural two results now have a completed Foundation successor, followed by Mathematics, Information Science, Classical Computation, Quantum Computation, Chemistry and Materials papers; the expanded Physics manuscript is the next unreleased paper.	This methods update reports the platform's growth without retroactively importing later answers into the two founding proofs. Each branch remains independently versioned, falsifiable and open to lawful extension.

1. Why this platform exists

Science is strongest when a result can survive contact with an unfavourable test by a person who owes the author, institution and prevailing consensus nothing. That ideal is not identical to the machinery by which modern research is funded, selected, published and amplified.

The concern is empirical, not rhetorical. A 2018 systematic review and meta-analysis of 75 studies found that industry-sponsored drug and device studies more often reported favorable efficacy results and favorable conclusions than studies with other sponsorship, while a scoping review found evidence that commercial sponsorship can shape which research questions are pursued (Lundh et al., 2018; Fabbri et al., 2018). These associations do not prove misconduct in any particular study or imply that public funding is neutral. They establish that funding provenance is a scientifically relevant variable and that disclosure alone is not the same as structural independence.

Selection systems also carry uncertainty. Empirical work on grant review has reported poor agreement and sensitivity to panel composition; a study of 3,156 Canadian health-research applications found ranking more reliable than rating, while both methods disadvantaged early-career applicants in that dataset (Gallo et al., 2023). A Cochrane review concluded that evidence about whether grant peer review improves funded research quality was limited and called for experimental study (Demicheli and Di Pietrantonj). Expert review remains indispensable; the evidence means it should be exposed to audit rather than treated as an infallible oracle.

The published record is not a neutral census of attempted science. Null and negative results are systematically underrepresented in many fields, which can inflate effects and distort reviews. Recent consensus work calls this a long-standing problem requiring action from funders, institutions, publishers and researchers (Nature Communications, 2025). The 2021 UNESCO Recommendation on Open Science likewise calls for methods, software, source code and outputs to be available for rigorous scrutiny; it specifically warns that paywalls and high article-processing charges can produce global inequalities (UNESCO, 2021).

Prediction introduces a newer version of the same epistemic problem. Opaque models can be useful instruments and can establish bounded predictive reliability under proper blind testing. They do not, by accuracy alone, expose a derivation from premise to law. NIST identifies validity, reliability, accountability, transparency, explainability and interpretability as related features of trustworthy AI (NIST AI RMF 1.0). Life-science researchers have similarly argued that data, code, model and workflow availability are necessary for third-party computational reproducibility (Heil et al., 2021).

Ernos Labs therefore rejects three substitutions:

1. funding success is not evidential closure;
2. professional consensus is not generated uniqueness; and
3. a black-box score is not an explicit scientific law.

This is not an argument against funding, expertise, consensus as a provisional coordination tool, or machine learning as an instrument. It is an argument that none of them may silently select the law later presented as fundamental. SFT's answer is a public, executable admission boundary: show the derivation, generate the alternatives, preserve the failures, expose the controls, bind the artifacts and let anyone reproduce acceptance or rejection.

The editorial conclusion is stronger than a request for incremental access. Knowledge institutions operating through competitive capital, scarce grants, prestige markets, subscription access and author charges inherit incentives that are not identical to free human inquiry. Even without individual bad faith, those mechanisms decide who can spend time asking questions, which questions appear fundable, what enters the visible record and who may read it. That is a direct scientific loss: excluded people do not merely lose access to known answers; humanity loses the questions and constructions they were never resourced or authorised to pursue. The documented funding, review, publication and access effects above are mechanisms by which that loss can occur.

Maria Smith develops SFT outside conventional academic credential and funding routes. In her editorial account, institutions would not have been expected to commission, credential or amplify this work before it existed. That fact is not offered as evidence that SFT is correct. It is an indictment of an admission system whenever a person's status determines whether the work may be heard before the work can be inspected. The remedy proposed here is equally demanding of insiders and outsiders: the derivation, complete census, controls and empirical record-not the biography-must speak.

2. Scientific ethos

The platform is built around six commitments.

2.1 Evidence before status

Credentials help identify relevant expertise and reviewers. They do not override a failed hash, an incomplete grammar, a missing control or an unfavourable measurement. Conversely, a mechanically accepted package does not manufacture specialist empirical meaning. Both mechanical and subject review are required at their proper boundaries.

2.2 Open access includes the failed rows

Source publication without candidate generation, rejected alternatives and negative controls is not enough. Every admitted SFT claim must retain favourable, unfavourable, failed and deliberately tampered results. A later contradiction is part of the record, not an inconvenience to be edited away.

2.3 Derivation and prediction are different evidence

A mathematical proof, computational construction, blind prediction and post-seal measurement are four distinct objects. The platform does not allow one to impersonate another. A proof cannot replace observation where nature is at issue; predictive accuracy cannot replace an unstated law.

2.4 Laws do not come from applications

Fold Protein, Fold Chess, Fold Go and Unison AI are future test and translation domains. They may falsify, validate or demonstrate laws only after the relevant main-corpus branches close. They cannot tune the fundamental theory. Their rebuilds are explicitly frontier work and are excluded from this release.

2.5 Authority is inspectable and revocable

Maria Smith is the scientific author and publication authority. The model's scientific admission authority is narrower: an accepted, reproducible engine receipt. If a public reproducer exposes a broken dependency, incomplete census, failed control, seal mismatch or empirical failure, the engine must reject the package regardless of authorship.

2.6 Open science is a practice, not a slogan

The repository is readable without Docker, cloud services or paid software. The proof kernel uses Python's standard library. Schemas, source, claims, receipts, tests, paper and prior-work ledger live together. The intended result is not trust in Ernos Labs. It is the ability to check Ernos Labs.

3. Exact scope of Smithian Fold Theory v3

This repository is the accessible third clean-room reconstruction of SFT. It is organised as a scalable tree intended eventually to address mathematics, information, computation, quantum computation, physics, chemistry, materials, biology, Earth and environmental science, astronomy, cosmology and disciplined engineering translation.

That programme remains open-ended, but the present versioned record now includes separately admitted Foundation, Mathematics, Information Science, Classical Computation, Quantum Computation, Chemistry and Materials branches and a locally prepared expanded Physics manuscript. This methods paper does not absorb those results: it records the common admission constitution. Earlier SFT observations may define questions and external correspondence only after a V3 claim seals; they cannot be copied as executable premises or answer tables.

The SFT derivational domain is presently governed by these constraints:

- one premise-free root theorem;
- no axioms in a registered derivation;
- no fitted, trained, free or target-derived parameters selecting a law;
- no semantic numerical zero as an SFT mathematical object;
- no negative or irrational values as foundational SFT quantities;
- no floating-point value as a proof value;
- no completed infinity or ungenerated continuum as a premise;
- exact positive rational parts only after they are themselves derived;
- structural held labels rather than imported signed quantities;
- generated candidate closure and forced assembled-form uniqueness; and
- explicit conditional status whenever a grammar is bounded and lacks a depth-independent certificate.

Computers necessarily use implementation indices, empty containers, Boolean flags and process status codes. These host mechanics are quarantined: they may implement a check but may not become SFT mathematical evidence. Python is the accessible host language, not a source of law.

4. Supersession and the clean-room rule

The project has grown beyond the organisation of its earlier generations. V3 exists to prove a new, independently executable reconstruction and to scale the corpus by scientific branch. The previous repositories and papers are **superseded as active theory authorities while the rebuild is in progress**. They remain immutable historical provenance and are not described as retracted or erased.

Prior work	Persistent reference	V3 treatment
SFT v2 main corpus	GitHub repository, comparison commit `0af7c4c26308d8ddb659a518ac52e2db5ea5dc82`	Preserved and hash-ledgered; never a v3 proof dependency
There Is No Nothing foundation paper generation	Zenodo concept DOI	Historical foundation correspondence; replaced only by a completed v3 foundation-branch paper
After Turing: The Fold Machine	Zenodo DOI	Historical computational-science corpus; to be replaced after the v3 computation branches close
Fold Protein paper	Zenodo record 21493135	Historical application evidence and empirical-method precedent; application rebuild deferred
No Dice and *Why There Is Uncertainty*	No Dice; Uncertainty	Historical quantum correspondence; no v3 quantum claim imported

The machine-readable [prior-work ledger](prior-work-ledger/manifest.json) binds the v2 corpus identity and selected source hashes. Opening a comparison entry is permitted only after the corresponding v3 result seals. Agreement, representation-equivalence, stronger or weaker closure, and disagreement are all reportable. V3 may not be back-fitted to manufacture agreement.

V3 uses Python for reach and legibility. A registered future v4 will be a fourth clean reconstruction in SFT-derived languages, compiler principles and runtime structures. V4 cannot retroactively author v3 laws. See the [self-hosting charter](docs/V4_SELF_HOSTED_REBUILD.md).

5. The single admission engine

All model admission passes through one versioned `SFTAdmissionEngine`. Drafts may exist outside it; laws may not. The engine is fail-closed: absence of any required object produces a rejection receipt and stops admission.

5.1 Admission stages

Stage	Required evidence	Failure consequence
Registration	Exact statement, one root route, admitted dependencies, provenance, no axioms, no free parameters, source identity	Reject before candidate execution
Enumeration	Generator, boundary, expected cardinality, duplicate-free candidate census, completeness certificate	Reject incomplete or hand-picked space
Forcing	One decision per candidate and exactly one survivor	Reject ambiguity or no result
Form closure	Minimality, named-shape uniqueness and boundary or generality certificate	Conditional status or rejection
Controls	False-premise, tampered-source, tampered-artifact and boundary controls	Reject if failure is not detected
Independent check	A deliberately distinct implementation regenerates the result and controls	Reject stored-answer or implementation-only agreement
Empirical extension	Preregistration, capability closure, separate target custody, seal-before-target, post-seal metrics, falsification and complete rows	Reject target leakage, opacity or selective retention
Census admission	Accepted receipt matching source and ordered manifest	No direct prose or file edit can admit a law

The engine checks a declared generator; it cannot magically prove that an author's grammar is complete. Every claim must therefore state its exact grammar boundary and provide a claim-specific completeness certificate. A finite census remains conditional unless its generality beyond that census is proved.

5.2 Exact arithmetic and prohibited answer flow

The kernel represents proof quantities with exact structures. Floating-point values are refused as proof objects. Measured decimal data may enter a separate empirical adapter, but never feed backward into law selection. Registrations explicitly list excluded inputs such as prior-source answers, benchmark targets, trained weights and application rules.

5.3 Receipts and cryptographic binding

Canonical serialization binds registrations, generated candidates, decisions, controls, closure evidence, validators and source manifests to SHA-256 identities. Hashing does not prove a theorem. It makes silent post hoc change detectable and identifies exactly which scientific object a receipt concerns.

An accepted receipt can be recomputed; a rejected receipt names the failed gate. The census accepts receipt-derived state only. Manually changing a status file does not change model authority.

5.4 Blind empirical execution

No v3 natural-science law has yet reached empirical admission. The engine already enforces the two-phase envelope required for future work:

1. register inputs, source, law, candidate space, target identity, metrics, controls and falsification conditions;
2. run prediction without target data or comparison code;
3. deny filesystem, network, subprocess, clock, environment, dynamic-import and foreign-function operations in the future capability-closed Fold language;
4. seal all predictions and traces;
5. have a separate custodian release the target only after the seal;
6. measure only preregistered quantities; and

7. preserve all rows, including failures and tampered controls.

Python's standard library cannot truthfully provide a universal operating-system sandbox for arbitrary native code. The repository therefore does not claim that it can. Official empirical prediction awaits a derived capability-closed Fold interpreter plus custody tests on macOS, Windows and Linux. This explicit limit is part of the result.

5.5 What machine checking does and does not establish

Machine checking establishes that the declared objects and algorithms satisfy the implemented constitutional gates. It does not prove that software contains no possible defect, that an incomplete scientific inventory is complete, or that formal consistency alone describes nature. Natural claims still require sealed empirical tests and specialist scrutiny. Independent human and cross-language replication remain open obligations.

6. Results of the third clean-room reconstruction

6.1 Result I - premise-free operational root theorem

Registered statement. No admissible operational statement, denial, record, proof object or derivational object is nothing: absence presents no counterexample, while every presented counterexample is an occurrence.

The result begins with no premise, axiom, dependency, parameter, earlier-corpus answer or application result. It generates the complete partition induced by presentation at the proof boundary:

Candidate	Exact operational form	Decision
`unpresented-absence`	No statement, denial, record, trace or identity is presented	Does not supply a counterexample object
`presented-occurrence`	A statement, denial, record, trace or identity is presented	Survives: presentation is an occurrence and therefore is not nothing

The partition is exhaustive at the declared boundary. A purported challenge either supplies an identifiable presentation or supplies no challenge artifact. The conclusion does not depend on the challenge's content, notation, length or internal depth. It is therefore certified depth-independent for operational statements, denials, records and proof challenges.

Minimality holds because deleting the presented witness deletes the only admissible proof object. Named-shape uniqueness holds because only the `presented-occurrence` class can enter execution, and its occurrence prevents it from instantiating nothing.

This is deliberately not advertised as a machine proof about an inaccessible, unexpressed metaphysical domain. The theorem closes the admissible operational boundary required to begin a derivation. That boundary discipline prevents a formal result from acquiring a larger philosophical claim by rhetoric.

Controls. The engine rejects an unpresented absence as a counterexample, detects changed source identity, detects a tampered two-survivor artifact, and refuses an uncheckable metaphysical broadening.

Evidence identities. Derivation seal `sha256:19c2e2af082dc9777c13bafb3bc561c7fa5cb96a51905fb26cfa06552ebf998f`; distinct-implementation validation `sha256:7942e78a54a932551ad6f5a2ceb8c5fb9122dcae0f18510052d402f42c604e3b`; current admission receipt `sha256:711864171e4d3a2f2734f0c2890965bcd81a0228349538751a3c80699c27d669`.

6.2 Result II - the structural One

Once the root occurrence is admitted, the next question is not "what number should be called one?" It is: what is the minimal representation that retains the admitted occurrence without omission or unforced addition?

The generator forms the product of three exhaustive root-coverage relations (`none`, `proper`, `complete`) and two addition relations (`no-extra`, `has-extra`). This yields six candidates:

Root coverage	Addition	Named form	Decision
none	no extra	`omitted-root`	Reject: does not retain the root
none	has extra	`replacement-extra`	Reject: does not retain the root
proper	no extra	`fragmented-root`	Reject: presupposes a cut and loses complete identity
proper	has extra	`fragment-plus-extra`	Reject: incomplete and added

Root coverage	Addition	Named form	Decision
complete	no extra	`exact-self-whole`	**Unique survivor**
complete	has extra	`whole-plus-extra`	Reject: imports material absent from the sole dependency

`exact-self-whole` is forced as the structural One. Removing retained identity makes it incomplete; adding material introduces an unforced dependency. Because the product classifies all representations solely by root coverage and addition, independently of the root's internal content or depth, the result is depth-independent at this declared boundary.

The admitted meaning is only structural self-wholeness. V3 has not yet derived the arithmetic numeral one, parts, division, a numerical domain, Fold fibres or subsequent laws. The boundary control specifically rejects those premature interpretations.

Evidence identities. Derivation seal `sha256:c62bb00cc30970ecf013bbc8225547a09395cf0719d3525f2a69bca1d582da7b`; distinct-implementation validation `sha256:ceaf767421b3450a862b10a6962da027c0467773a2e1be5115f554d2c3356083`; current admission receipt `sha256:68332624c276dc5d0ddfd568a26b95e20d7d5665fcdb7e825ff1b7b19bf1d51c`.

6.3 Current quantitative verification record

At this release's source state, the public route reports:

- 2 registered and model-admitted derivations;
- 60 unit, adverse-control and end-to-end tests;
- 15 core engine modules;
- 1,264 of 1,264 core-engine executable lines observed by the standard-library tracer; and
- a fresh ordered rerun of both derivations and their distinct validators.

"100%" means executable-line coverage of the named core engine package. It does not mean 100% path coverage, proof of semantic correctness, absence of every future defect or completion of the SFT programme.

The current independent validators are implementation-distinct Python routes, not yet independent research teams or a foreign-language replication. The repository calls that status `independently_replicated` within its mechanical schema, while this paper states the narrower empirical fact to prevent overreading.

7. One public verification route

From the repository root, run the interpreter available on the host:

```
python3 -m sft verify-all   # macOS and most Linux systems
python -m sft verify-all    # Python installations exposing `python`
py -m sft verify-all        # standard Windows Python launcher
```

These are operating-system spellings of one engine entry point, `sft verify-all`. The command performs repository and policy validation, all unit and end-to-end checks, exact core-engine coverage, ordered manifest loading, fresh derivation execution, control execution, distinct-validator execution and immutable receipt comparison. Any missing claim, changed source, dependency-order error, failed control, uncovered engine line or receipt drift returns failure.

The project has no runtime dependency outside the Python standard library and requires no Docker, virtual machine, network service or paid account. Publication tooling used to render the archival PDF is not part of derivational authority.

The continuous-integration matrix is configured for macOS, Windows and Ubuntu with supported Python versions. A green hosted matrix is portability evidence, not a source of scientific law.

8. Repository as a knowledge tree

Path	Scientific responsibility
[`sft/foundation`](sft/foundation/)	Root theorem, One, parts, Fold and future form grammar
[`sft/mathematics`](sft/mathematics/)	Exact arithmetic, discrete structures, algebra, order, geometry, topology, probability, dynamics, logic and composition
[`sft/information_science`](sft/information_science/)	Distinguishability, encoding, uncertainty, channels, coding and correspondence
[`sft/computation`](sft/computation/)	Formal computation, computability, complexity, algorithms, semantics, distributed systems, security, learning and scientific computation
[`sft/quantum_computation`](sft/quantum_computation/)	Reversible and quantum models, circuits, communication, correction, verification and limits
[`sft/physics`](sft/physics/)	Physical laws and sealed tests
[`sft/chemistry_materials`](sft/chemistry_materials/)	Chemical and material structure
[`sft/biology`](sft/biology/)	Biological structure, genetics, evolution and living systems
[`sft/earth_environment`](sft/earth_environment/)	Earth, climate, geology, oceans and environmental systems
[`sft/astronomy_cosmology`](sft/astronomy_cosmology/)	Astronomy, cosmology and observation
[`claims`](claims/)	Complete claim packages: registrations, censuses, decisions, controls and certificates
[`receipts`](receipts/)	Immutable accepted, rejected and prebaseline execution records
[`census`](census/)	Ordered model authority and branch inventory
[`experiments`](experiments/)	Registrations, sealed predictions, post-seal measurements and adverse controls
[`correspondence`](correspondence/)	Post-derivation comparison with established theories
[`prior-work-ledger`](prior-work-ledger/)	Hash-bound v1/v2 provenance with no derivational authority
[`publications`](publications/)	Methods paper, completed branch papers and final synthesis protocols

See the full [directory map](docs/DIRECTORY_MAP.md), [claim lifecycle](docs/CLAIM_LIFECYCLE.md), [engine authority](docs/ENGINE_AUTHORITY.md) and [empirical method](docs/EMPIRICAL_METHOD.md).

9. Publication constitution and roadmap

This inaugural paper publishes the method, root result, structural One and platform needed to audit future work. It does not waive the stricter branch paper rule.

For each science branch:

1. freeze a current-knowledge obligation inventory;
2. derive every included law in dependency order;
3. pass each package through the single engine;
4. complete formal, computational and empirical validation appropriate to the claim;
5. preserve every boundary and unfavourable result;
6. produce a comprehensive evidence map from every paper statement to source, candidate census, decision, receipt and artifact hash; and
7. publish a full branch paper only when the branch publication gate closes.

No summary paper may conceal unclosed obligations. The final rewritten Theory of Everything paper will be produced only after every frozen branch has a completed branch paper and the global census passes cross-branch consistency and evidence-map checks. Unresolved external questions must remain named frontiers.

The immediate v3 programme is to continue the foundation from the structural One to parts, the exact derivational domain, Fold fibres and the generated Fold grammar. Mathematics and scientific branches follow only as dependencies are admitted. Application translations and v4 self-hosting remain frontier work.

10. Participation, review and the Ernos Labs designation

This release opens the source, evidence and challenge route. Formal external contribution admission will open after the foundation interfaces and governance release are stabilized; early proposals, falsifications and reproductions are welcome now through email, Discord and GitHub.

Anyone may download, inspect, run, fork, cite, criticize and attempt to invalidate the work under the published licenses. Maria Smith retains copyright and authorship. The **Ernos Labs** name is separately reserved as a conformance designation. A project may describe itself as Ernos Labs only while it:

- publishes the applicable constitution and full dependency chain;
- excludes target answers and fitted parameters from law selection;
- generates and retains its candidate space and rejected alternatives;
- publishes source, controls, receipts and reproducible evidence;
- distinguishes theorem, finite computation, empirical result and frontier;
- preserves unfavourable outcomes; and
- accepts good-faith technical review.

The designation does not prevent independent forks or criticism and does not convert disagreement into infringement. It prevents an open license from being misrepresented as scientific endorsement. See the [designation policy](TRADEMARK_POLICY.md), [licensing map](LICENSING.md) and [contribution guide](CONTRIBUTING.md).

Review should identify the exact claim, premise, dependency, generator, candidate, decision, artifact, metric, boundary or falsification at issue. Personal status is neither a proof nor a refutation.

11. Limitations and falsification routes

This paper's strongest claims are narrow by design.

- The operational root theorem does not certify an unexpressed metaphysical domain.
- The structural One is not yet arithmetic one and does not import parts or Fold structure.
- The two claims owned by this methods paper do not replace the separately published Foundation branch.
- No v3 natural-science law has passed a sealed empirical run.
- The independent checks are implementation-distinct, not yet institution-independent or cross-language replications.
- Core-engine line coverage does not establish defect absence.
- Hash integrity does not establish mathematical truth.
- The engine validates the declared grammar; scientific review must still challenge whether that boundary and completeness certificate are adequate.
- Hosted cross-platform CI must pass after public push before platform portability is described as externally observed.

The work can be invalidated or narrowed by producing, among other things: an operational counterexample outside the root presentation partition; a missing coverage/addition class for structural One; a second surviving form inside the declared grammar; a source or receipt mismatch; an adverse control that the engine accepts; an undeclared answer-producing dependency; target leakage; or a registered empirical failure. A valid challenge is evidence, not hostility.

12. Formal evidence model

The prose above describes the scientific policy. This section specifies the actual objects consumed and produced by engine version `sft-v3-single-admission-engine/1`.

12.1 Registered claim object

A `ClaimRegistration` is immutable and contains:

Field	Type and requirement	Scientific function
`claim_id`	nonempty exact label	Stable identity across source, census and receipt
`title`, `branch`, `statement`	nonempty text	Human-readable scope; never substitutes for execution
`evidence_mode`	`formal` or `empirical`	Selects the appropriate external gate before execution
`root_theorems`	empty only for the root; otherwise exactly the one admitted root	Prevents hidden alternative foundations
`dependencies`	ordered, duplicate-free admitted claim IDs	Makes the derivation graph executable and acyclic in census order
`axioms`	must be empty	Enforces the no-axiom constitution
`free_parameters`	must be empty	Excludes fitted, learned or chosen answer-producing quantities
`provenance`	one or more allowed classes	Separates direct forcing, forward forcing, constitutional relation and observational derivation
`source_hash`	canonical SHA-256 identity	Binds registration to the files executed by the official runner

The engine separately computes an executed source manifest from the actual loaded files and refuses a mismatch with `source_hash`. A registered hash is therefore not accepted merely because it has SHA-256 syntax.

12.2 Candidate and decision objects

The generator returns a `CandidateCensus` containing a rule, exact grammar boundary, positive expected cardinality, completeness-certificate hash and ordered candidate tuple. Each `Candidate` has a unique identifier, exact form and trace hash. The engine rejects a missing rule, missing boundary, nonpositive cardinality, count mismatch, duplicate identifier, unnamed form, invalid trace hash or absent completeness identity.

Candidate decisions are not supplied as an unbound result table. The engine calls `decide_candidate` exactly once for each generated item in generated order. Each returned `CandidateDecision` must name that candidate, state whether it survives, give an explicit reason and supply a proof hash. The decision IDs must equal the candidate IDs one-for-one and in order. Exactly one `survives` field must be true.

12.3 Closure, controls and seals

`ClosureEvidence` records one of three scopes:

- `conditional_grammar`: evidence is preserved but barred from model dependency;
- `finite_complete`: the complete declared finite space is closed; or
- `depth_independent`: the exact boundary is closed without dependence on an enumerated depth and requires a valid generality certificate.

Every scope requires an exact boundary, successful minimality, successful named-shape uniqueness and a closure proof hash. The four mandatory `ControlResult` objects are `false_premise`, `tampered_source`, `tampered_artifact` and `boundary`. Each carries expected behaviour, observed behaviour, pass state and receipt hash. Missing, duplicate, failed or malformed controls halt the claim.

The derivation seal is the canonical SHA-256 identity of this exact tuple:

```
{
  engine_id,
  registration,
  candidate_census,
  ordered_decisions,
  closure_evidence,
  ordered_controls
}
```

Canonicalization sorts mapping keys, preserves list and tuple order, converts typed records to field maps, serializes enum values by their exact labels and represents a rational host value as its numerator and denominator. Hashes bind identity and ordering; they do not substitute for the substantive checks.

12.4 Distinct validation and final receipt

The validator receives the sealed derivation and must return its own validator identity, implementation hash, the exact seal it checked, certificate hash, recomputation declaration and pass state. The engine rejects a missing validator, malformed identity, implementation identity equal to the registered source, wrong seal, stored-answer declaration or failed result.

For empirical mode, an additional `EmpiricalValidation` binds the experiment registration, capability certificate, target-custody certificate, evaluator seal check, target release order, all-row retention, source identities, measurements, falsification condition and measurement receipt. Formal mode rejects an unregistered empirical validator; empirical mode rejects its absence.

The final `EngineReceipt` contains the engine and claim identities, acceptance and model-admission booleans, closure and external statuses, halted stage, violations, every ordered gate result, derivation seal, independent-validation identity, optional empirical identity and a receipt hash over every preceding field. The receipt hash itself is excluded while calculating that identity.

13. Executable engine algorithm and authority flow

The engine algorithm can be stated without implementation-specific syntax:

```
RUN(program, distinct_validator, optional_empirical_validator):
  registration := program.registration
  REQUIRE registration is exact, source-bound, axiom-free,
          parameter-free and dependency-admitted

  census := program.generate_candidates()
  REQUIRE census matches its positive cardinality, unique identities,
          exact forms, trace identities and completeness certificate

  FOR candidate IN census.generated_order:
      decisions.append(program.decide_candidate(candidate))
  REQUIRE decisions correspond one-for-one with census
          and exactly one candidate survives

  closure := program.closure_evidence(decisions)
  REQUIRE exact boundary, minimality and named-shape uniqueness
  IF closure is depth-independent:
      REQUIRE generality certificate

  controls := program.run_controls()
  REQUIRE exactly one passing result for every mandatory control kind

  seal := HASH(engine, registration, census, decisions, closure, controls)
  external := distinct_validator.recompute(seal, declared_inputs)
  REQUIRE distinct implementation, correct seal, valid certificate and pass

  IF registration is empirical:
      REQUIRE preregistered, isolated and custodially separated measurement
  ELSE:
      REQUIRE no empirical validator was silently added

  admitted := closure in {finite-complete, depth-independent}
  EMIT accepted receipt
  IF admitted:
      authority_ledger.admit(receipt)
```

Every `REQUIRE` is fail-closed. An exception raised by a claim callback is converted into a rejected receipt at the current gate. Later callbacks are not executed. Rejected and conditional evidence can be preserved, but the authority ledger accepts only an internally consistent receipt with both `accepted_evidence=true` and `model_admitted=true`.

The repository coordinator enforces the second authority boundary. It builds the source manifest, runs the engine, writes the receipt atomically, verifies the stored representation and only then permits a census update. When rebuilding authority, it reads the ordered census, verifies each stored receipt, checks dependency order and reconstructs the ledger from accepted receipts. A direct edit to `census/claims.json` cannot forge a working dependency because the next complete verification reruns the derivation and compares the recomputed receipt object with both the census hash and immutable receipt file.

13.1 Engine module responsibilities

Module	Responsibility and enforced boundary
`authority.py`	In-memory dependency authority populated only by admitted receipts; rejects forgery and conflict
`canonical.py`	Deterministic serialization and SHA-256 identities, including exact rational representation
`empirical.py`	Two-phase prediction envelope; refuses target access before seal
`engine.py`	Ordered fail-closed orchestration, violation checks and receipt construction
`errors.py`	Carries the complete rejected receipt when execution halts
`exact.py`	Refuses Boolean-as-count, nonpositive count, semantic zero, negative values, beyond-One parts and floating proof quantities
`external.py`	Separate-process validator adapter with source-manifest identity and strict result parsing
`isolation.py`	Capability and target-custody certificates without pretending Python is a universal sandbox
`model.py`	Immutable typed evidence records and derivation/validator protocols
`portable.py`	One macOS/Windows/Linux host vocabulary and a restricted child-process environment
`publication.py`	Branch-inventory and final-paper completeness gates
`receipt_io.py`	Atomic receipt persistence, exact-field parsing and hash verification
`repository.py`	Sole receipt-to-census path and authority reconstruction
`source.py`	Repository-contained source artifact hashing; rejects missing, empty or escaping files
`__init__.py`	Narrow public interface for the one engine package

14. Complete executable trace: the operational root theorem

This section reproduces the full public evidence chain for `SFT-ROOT-THERE-IS-NO-NOTHING`. The authoritative machine objects remain the JSON and Python artifacts named in the evidence map; the table below makes that chain readable without asking the reader to infer missing steps.

14.1 Registration trace

Property	Registered value
Claim	`SFT-ROOT-THERE-IS-NO-NOTHING`
Title	Premise-free operational root theorem
Branch	foundation
Evidence mode	formal
Statement	No admissible operational statement, denial, record, proof object or derivational object is nothing: absence presents no counterexample, while every presented counterexample is an occurrence.
Root theorems	empty tuple
Dependencies	empty tuple
Axioms	empty tuple
Free parameters	empty tuple
Provenance	`direct_forcing`
Excluded inputs	v2 source/certificates/answers; conventional ontology or logic as premise; application-selected rules; empirical targets
Source manifest	`sha256:c3de6a8fc34d927dce243992a5a0de610f0c8701084ae08b3c9828838f75d915`

The root exception is precise: every downstream claim must trace through exactly this one admitted root, while this claim must cite no root and no dependency. A denial is not imported as a logical axiom. It is submitted as an operational challenge to the generator.

14.2 Generator and completeness trace

Generation rule: partition every purported operational counterexample by whether it emits an identifiable presentation at the proof boundary.

Exact boundary: all operational statements, denials, records and proof challenges, independent of internal length or notation.

The classifier is a single property: presentation at the proof boundary. If no statement, denial, record, trace or identity is emitted, no counterexample object enters the proof boundary. If one is emitted, the emitted object is an occurrence. There is no third operational class relative to this classifier: any purported third challenge either emits an identifiable presentation and is in the second class, or does not and supplies no proof artifact.

Expected cardinality is therefore the counted value two. The completeness certificate is `sha256:15bdb6e089afecc0a45498eabf39d25a1c93e2f5de8346fe998860bde750ed9d`.

14.3 Generated candidates and decisions

Candidate	Exact form	Candidate trace	Decision and reason	Decision proof
`unpresented-absence`	No statement, denial, record, trace or identity is presented; there is therefore no operational counterexample to admit.	`sha256:8b0f607cd624d7e45fb5af28484d0f9ae12f56636c09e7d0b72730f572a81d0a`	Reject. Without a presentation there is no counterexample object to admit.	`sha256:a05b0ae640948852199f68e8042506212c38e6d0c3bdc72a1c1c870029626eef`
`presented-occurrence`	A statement, denial, record, trace or identity is presented; that presentation is an occurrence and therefore is not nothing.	`sha256:6cb18c5ee96566de7b9e9a08633e48be1e3dd7590295920858889f67fee04222`	**Survives.** Presentation supplies the witnessed occurrence required by the theorem.	`sha256:c6b4d78467ea08f053d781f0f024c4fbc5960b53eb8f4f044ebdb20811156044`

Exactly one candidate survives. The conclusion is not selected by a target value; it follows from applying the same presentation rule to every member of the generated partition.

14.4 Minimality, uniqueness and generality

Minimality: removing the presented witness removes the only admissible proof object. What remains is not a smaller counterexample; it is the absence of a submitted challenge. Named-shape uniqueness: within the exhaustive presentation partition, only `presented-occurrence` enters execution, and its status as an occurrence prevents it from satisfying the proposed nothing class.

The reasoning depends only on presentation, not on content, notation, length, host encoding or internal depth. The exact boundary is therefore certified `depth_independent`. Closure proof: `sha256:b06285bf05b81202ccd4331e50c6d5e4f32db07d4f14aecbcb38ac8825ac7716`. Generality certificate: `sha256:67bee220b6f1c353142b2021a3c83640c1ed69235429155e6ebdf3bc1ee5a608`.

14.5 Root adverse controls

Control	Deliberate challenge	Expected and observed result	Receipt identity
False premise	Treat unpresented absence as a presented counterexample	Expected rejection because no identity or trace exists; observed `unpresented-absence` not admissible	`sha256:afe5d206d2c2de5d7ec6459854a1a4dec09310b2b3514f9569325262c70090a7`
Tampered source	Alter registered derivation source identity	Expected rejection before enumeration; observed changed identity differs	`sha256:4c7f233ed8e6c655aaf1094f2123f8642425ea92e9b9176dc8598b4adaccbc36`
Tampered artifact	Mark both generated classes as surviving	Expected rejection for nonunique forcing; observed two-survivor set	`sha256:a4a220c6532648805adcc82501fa683fc4c87572af689a0a9022abd97255fc0a`
Boundary	Assert an unexpressed metaphysical object outside the operational boundary	Expected refusal because no checkable object is supplied; theorem remains operationally scoped	`sha256:8d3a9ef27fade730c4c3484657e789118f8f2a67045e21fecc093da702fad8c0`

14.6 Root seal, distinct recomputation and admission

The engine seals registration, two generated candidates, two decisions, closure and four controls as `sha256:19c2e2af082dc9777c13bafb3bc561c7fa5cb96a51905fb26cfa06552ebf998f`. The separate validator implementation identity is `sha256:54ec0bbe6fecc15920c05fc3f75df6a66957c3c15a4076245ab22c36a7b766ca`. It regenerates the presentation partition from declared inputs, recomputes the survivor and controls, and produces validation identity `sha256:7942e78a54a932551ad6f5a2ceb8c5fb9122dcae0f18510052d402f42c604e3b`.

All eight receipt gates pass: registration, enumeration, forcing, form closure, controls, seal, distinct validation and model admission. The final receipt is sha256:711864171e4d3a2f2734f0c2890965bcd81a0228349538751a3c80699c27d669. It contains no empirical validation because the registered claim is formal.

15. Complete executable trace: the structural One

SFT-FOUNDATION-ONE-001 is the first forward-forced result. It can run only after the root receipt has entered a fresh in-memory authority ledger.

15.1 Registration trace

Property	Registered value
Claim	SFT-FOUNDATION-ONE-001
Title	The structural One
Branch	foundation
Evidence mode	formal
Statement	The unique minimal representation of the admitted root occurrence is its complete self-whole with no unforced addition; this structural identity is the One.
Root trace	exactly SFT-ROOT-THERE-IS-NO-NOTHING
Dependencies	exactly SFT-ROOT-THERE-IS-NO-NOTHING
Axioms	empty tuple
Free parameters	empty tuple
Provenance	forward_forcing
Excluded inputs	v2 One definition/implementation; conventional numerical one; application-selected identity; measured magnitude
Source manifest	sha256:6ae512f49a7053affbd2587a859679ebc636a039a3e1d4a22f59a14f58c3a0f3

The result asks only how an admitted occurrence can be represented relative to retention of that occurrence and introduction of material not present in its sole dependency. It does not assume number, magnitude or arithmetic identity.

15.2 Generator and completeness trace

For root coverage, the generated relation has three exhaustive classes:

- none: none of the admitted identity is retained;
- proper: a strict fragment is retained, presupposing a cut; and
- complete: the admitted identity is retained as a whole.

For material beyond the root, the generated relation has two exhaustive classes: no-extra and has-extra. Their Cartesian product has counted cardinality six and exhausts representations classified solely by these two relations. The generator is not given the desired form name.

Completeness certificate:
sha256:4d733e780b0e83a927e6c34268290ee82101f0a1f790e523fceb2ca3b581d892. Exact boundary: all representations classified solely by coverage of the admitted root occurrence and addition of material not supplied by it.

15.3 Six generated forms and full decision trace

Candidate	Generated relation	Candidate trace	Decision and reason	Decision proof
`omitted-root`	none coverage; no extra	`sha256:ab3bf2a7990ce67f52063b78af5a48c211c4228cd408243dc087c2b275779d95`	Reject: does not retain the admitted root occurrence	`sha256:638e651d319bdea79d55c6c36849378872a99e0d48343c4b7caa6b85a93f151e`
`replacement-extra`	none coverage; has extra	`sha256:76f402b7417cebd506e7cee8818524ebfaed73e244dde2cc30cd7c3a551a22f6`	Reject: does not retain the admitted root occurrence	`sha256:c6493c8c7ae0776d313afd11f6d7846b26db8dc2ea597568210531258b387ca1`
`fragmented-root`	proper coverage; no extra	`sha256:8adaa4ecace7f1914f47b40903a895113c25d44f928513ca29701f8787fad689`	Reject: presupposes a cut and loses complete admitted identity	`sha256:dab5ba1c22dca023d5c38434a0987490c3d1bb9525095ad2975d74df0ecac403`
`fragment-plus-extra`	proper coverage; has extra	`sha256:9f299064dc7fc26084c0cbeeda5a672213bdc3369f3f8fc78f8cd3799518b30d`	Reject: presupposes a cut and loses complete admitted identity	`sha256:5481c32d5600ff812dc3460e9796e26be160ef39a9b2c0441dc6a563732a9472`
`exact-self-whole`	complete coverage; no extra	`sha256:258126da54beeac598231fbe1bed20681ee32cbdd5bd601cf8b60af4075e92ea`	**Survives:** retains the complete admitted occurrence without omission or addition	`sha256:80937d3b539845ece0e167e0cafc8e3220609d0cd25e7c5670c71c856baa83d0`
`whole-plus-extra`	complete coverage; has extra	`sha256:01e607731f9aeeabd723352913692372bccac11fa801d9de825b64e9a229137c`	Reject: introduces material absent from the sole admitted dependency	`sha256:4b4a458070f0eb421838e93d79667908579afaae995913e359f88cf66183a024`

Exactly one form is both complete and addition-free. Calling it the structural One records its relation to the admitted occurrence; it does not smuggle in a conventional numeral.

15.4 Minimality, uniqueness and generality

Deleting any retained root identity makes the candidate incomplete. Adding any material creates an occurrence not supplied by the sole dependency. Therefore `exact-self-whole` is minimal. Because every pair in the complete coverage and addition product was generated, and only `(complete, no-extra)` survives, the named shape is unique at the boundary.

The classifier does not inspect the root occurrence's content, notation, size or internal depth. It applies to an admitted identity as such. Closure is therefore depth-independent at the exact two-relation boundary. Closure proof: `sha256:87c7bd8ee1b561223c7d93309ebfd8662b89cff819f5a38b30f50dd5c48f5b27`. Generality certificate: `sha256:74e4c3761ec998027e1b12ed7c5f2581cd45132d11134cee15e13dffc8c64a19`.

15.5 Structural One adverse controls

Control	Deliberate challenge	Expected and observed result	Receipt identity
False premise	Substitute an omitted root as a representation	Expected rejection; observed `omitted-root` does not survive	`sha256:0dceced6a8585727559c36b83c2f4b0d0061703175b7ab66d4a653b3be67600a`
Tampered source	Change source identity	Expected source refusal; observed identity differs	`sha256:e4188d90cf7ccec55e447016899f3e85a391963b406e3ff5f914e99dcc3daaa3`
Tampered artifact	Add material to the complete root	Expected rejection; observed `whole-plus-extra` does not survive	`sha256:c857b9ae91fe5c805ba7e9d7b170f22ea1c25045938c3ae5c2599c5f948a199b`
Boundary	Promote the result to a numerical magnitude	Expected refusal; observed meaning remains structural self-wholeness only	`sha256:f9b13fb3dc82eeffcfa15517eb3320854a0b96759feb5e9846c0c352df66bce9`

15.6 One seal, distinct recomputation and admission

Derivation seal: `sha256:c62bb00cc30970ecf013bbc8225547a09395cf0719d3525f2a69bca1d582da7b`. Distinct validator implementation: `sha256:123f67f4797383feb1a659ec3b66812e0870e507a81ca0b061434bd273fad66b`. Validation identity: `sha256:ceaf767421b3450a862b10a6962da027c0467773a2e1be5115f554d2c3356083`. Final receipt: `sha256:68332624c276dc5d0ddfd568a26b95e20d7d5665fcdb7e825ff1b7b19bf1d51c`.

Registration, enumeration, forcing, form closure, controls, seal, distinct validation and model admission all pass. No empirical object is attached.

16. Verification and adverse-test census

The release suite contains 60 tests grouped by purpose rather than accumulated as an undifferentiated success number.

Test group	Count	Covered surfaces
Engine behaviour	29	formal and empirical acceptance, conditional exclusion, axioms, parameters, enumeration, uniqueness, controls, independent identity, target order, capability policy, exact domain, publication gates, census authority, source drift and external process recomputation
Engine totality and malformed surfaces	18	forged receipts, canonical/exact guards, every invalid registration class, callback exceptions, malformed census/closure/control/external/empirical objects, isolation/custody fields, source escape, parser strictness and conflicting receipts
Structural One	4	six-form product, sole survivor, admitted-root dependency and tampered second survivor
Root theorem	4	premise-free registration, two-class partition, complete engine execution and tampered survivor
Repository scaffold	5	census identity, generation state, fail-closed policy, JSON parseability and required guidance

The public verifier launches the full suite under Python's standard-library `trace` module. It scans every `sft.engine` source module, counts executable lines, and fails on any line marked missing or any module not executed. The current count is 1,264 executed of 1,264 executable lines across 15 modules. Temporary trace files are discarded after the run.

After software verification, the command loads `census/execution_manifest.json`. It requires the manifest claim order to match `census/claims.json` exactly. It then creates a blank authority ledger, loads each claim factory from its registered execution file, rebuilds the source manifest, reruns the engine and distinct validator, compares the entire newly computed receipt to the immutable stored receipt, and admits it to the fresh ledger before executing its dependants. No verification run writes scientific evidence.

The suite deliberately tests unfavourable paths, including:

- axiom or parameter present at registration;
- missing, duplicate or reordered candidates and decisions;
- no survivor or multiple survivors;
- failed or duplicate controls;
- forged, incomplete, extra-field or hash-mismatched receipts;
- same implementation presented as independent;
- validator subprocess failure or malformed output;
- source file missing, empty or outside the repository;
- target access before seal, wrong target identity and duplicate seal;
- forbidden prediction capability and mismatched custodian;
- conditional result attempting to become a dependency;
- branch paper with an open frontier; and
- final paper without every completed branch paper.

These controls demonstrate that named failures are detected. They do not prove that the test author has imagined every future failure mode.

17. Threat model, portability and empirical custody

17.1 Threats addressed

The engine is designed against silent answer import, post hoc result selection, incomplete candidate presentation, source drift, artifact tampering, forged status, dependency bypass, target leakage, selective reporting and publication before inventory completion. Each threat is mapped to an executable gate and a retained identity.

17.2 Threats not claimed solved

No hash system protects a compromised machine that changes both program and reported evidence before any honest copy exists. No Python process can universally sandbox hostile native code across all operating systems. No mechanical schema can decide whether a scientific grammar omitted a concept the author failed to generate. No internal validator equals independent community replication. These are disclosed review surfaces, not hidden assumptions.

The publication response is redundancy: public git history, DOI archives, checksums, separate-process validators, future generated-C or SFT-language certificates, independent forks and explicit empirical custody. A contradiction between copies becomes observable.

17.3 Cross-platform contract

The runtime is standard-library-first Python and recognizes exactly `macos`, `windows` and `linux` as receipt metadata. Host identity cannot select a scientific branch. Child validator processes receive only a controlled minimal environment rather than arbitrary parent variables. The same engine policy and candidate logic run on each platform.

The repository offers one logical verifier, `sft verify-all`; the interpreter launcher is spelled `python3`, `python` or `py` according to a host's Python installation. Docker, a virtual machine and cloud execution are not required.

17.4 Future official empirical run

An official natural-science prediction requires two custodians or equivalently separated roles. The prediction custodian receives the registered program, allowed inputs and only target identities. The capability-closed interpreter has no instruction for clock, environment, filesystem, network, subprocess, dynamic import or foreign function access. Its certificate binds interpreter, program, inputs, denied capabilities, trace and proof of target absence.

The target custodian binds the same experiment registration and target identities, retains target content, and produces a cross-bound certificate that the data were not released before the prediction seal. Only then may the evaluator verify the seal, open the target, calculate preregistered measurements and emit the complete-row receipt. A favourable score without those objects may support an instrument-performance report but cannot enter SFT as a closed law.

18. Full paper-to-evidence map

Every derivational statement in this paper maps to public artifacts. Relative paths are stable inside the repository and Zenodo bundle.

Paper subject	Registration and generator	Decisions and closure	Independent proof and receipt
Operational root theorem	`claims/SFT-ROOT-THERE-IS-NO-NOTHING/registration.json`; `candidate_census.json`; `sft/foundation/root_theorem.py`	`elimination_receipt.json`; `controls.json`; `STATUS.md`	`independent_validator.py`; `certificate.json`; `receipts/engine/model_admitted/SFT-ROOT-THERE-IS-NO-NOTHING-711864171e4d3a2f.json`
Structural One	`claims/SFT-FOUNDATION-ONE-001/registration.json`; `candidate_census.json`; `sft/foundation/one.py`	`elimination_receipt.json`; `controls.json`; `STATUS.md`	`independent_validator.py`; `certificate.json`; `receipts/engine/model_admitted/SFT-FOUNDATION-ONE-001-68332624c276dc5d.json`
Engine policy	`CONSTITUTION.md`; `governance/engine_policy.json`; schemas under `governance/`	`sft/engine/engine.py`; `model.py`; `exact.py`; `isolation.py`	`tests/test_engine.py`; `tests/test_engine_totality.py`; `docs/VERIFICATION.md`
Census authority	`census/execution_manifest.json`; `census/claims.json`	`sft/engine/repository.py`; `receipt_io.py`; `authority.py`	`sft/verification.py`; model-admitted receipt directory

Paper subject	Registration and generator	Decisions and closure	Independent proof and receipt
Empirical method	`docs/EMPIRICAL_METHOD.md`; experiment schema and registration template	`sft/engine/empirical.py`; `isolation.py`	Current state: infrastructure tested, no official v3 empirical claim admitted
Prior-work supersession	`prior-work-ledger/manifest.json`; `NOTICE.md`	Post-seal correspondence files	V2 commit and SHA-256 identities recorded in the ledger
Publication constitution	`publications/BRANCH_PAPER_PROTOCOL.md`; `FINAL_TOE_PAPER_PROTOCOL.md`	`sft/engine/publication.py`	Publication failure tests in engine suite
Rights and participation	`LICENSE`; `LICENSING.md`; `TRADEMARK_POLICY.md`; `CONTRIBUTING.md`	`governance/ERNOS_LABS.md`	Public repository history and archived DOI edition

The two `WHY_DERIVATION_CHECK.md` files provide a cold-readable bridge from the formal artifacts to human interpretation. They are guidance, not additional premises. `correspondence/foundation/` records historical comparison only after the v3 seals. The v2 source files recorded in the prior-work ledger have hashes `sha256:0bf79de6c3f42784424537481daf9df310c71cc1c502c148e6aa27bcd1f2a401` for `STANDARDS.md` and `sha256:42c4be709dcd9edcfbedc70ee82055a8660d9658de21758561fd46e068a727bf` for `OneFoldMaster.md`; neither is in a v3 dependency tuple.

19. Conclusion

Ernos Labs is an attempt to restore empirical primacy and public reproducibility at a scale large enough for an ambitious unification programme. It does not ask readers to accept SFT because of an institution, a credential, a consensus vote, a funding decision or an oracle's score. It asks them to inspect a finite claim, run its construction, attack its boundary, reproduce its receipt and preserve the outcome.

The present result is intentionally small: one premise-free operational root, one forced structural whole, one public admission engine and one open knowledge tree. Its significance, if any, will be earned only by whether the same method continues to close later branches without importing their answers-and whether independent researchers can prove where it succeeds or fails.

Data, code, licenses and citation

All source, candidate censuses, decisions, controls, certificates and receipts for this paper are contained in this repository and archival release bundle. Code and generated software artifacts are licensed under the Apache License 2.0. This paper, documentation and non-code evidence are licensed under Creative Commons Attribution 4.0 International. Maria Smith retains copyright. Third-party references retain their own rights.

Use [`CITATION.cff`](CITATION.cff) for software citation and cite this paper as Smith M. (2026), *There Is No Nothing: A Premise-Free Operational Foundation and an Open Verification Platform for Smithian Fold Theory*, doi:10.5281/zenodo.21627646. Cryptographic checksums are provided in the release bundle.

Foundation and full-field reconstruction roadmap - version 0.3.0

PUBLISHED SAME-PAPER SUCCESSOR VERSION 0.3.0. The preceding version 0.2.0 remains permanently preserved in the Zenodo version chain. This version publishes the shared two-layer roadmap governing the open reconstruction programme through Chemistry; Materials is not part of this release.

Methods Paper 00 owns the shared empirical constitution and publication lifecycle, not downstream laws. Version 0.3 adds the complete two-layer roadmap governing all fifteen branches. The two inaugural proofs and every existing receipt remain unchanged.

This amendment adds no scientific admission by prose. Every result already in the paper retains its exact receipt and evidence boundary. Every future result must pass the untouched engine independently. A branch described as complete to a dated inventory remains permanently open to lawful extension, correction, falsification and discoveries that satisfy the same public standard.

Purpose

This directory defines the dependency-ordered plan for every scientific branch in the Smithian Fold Theory open knowledge tree. It does not admit a law, change a receipt or declare a branch complete. Scientific status comes only from the protected admission engine, immutable receipts, the live categorical owner census and the applicable publication gate.

Each branch is developed in two layers.

Layer One - foundational reconstruction

The branch first establishes the smallest complete body of laws needed for the field to exist as its own science. This layer must:

1. freeze the branch boundary and dependencies;
2. audit every relevant V1/V2 statement and every current-knowledge subject;
3. assign each atomic obligation to exactly one categorical owner;
4. derive the branch's primitive objects, distinctions, operations, observations and composition laws from already admitted dependencies;
5. generate the declared candidate grammar and eliminate every alternative;
6. establish minimality, named-shape uniqueness and the exact finite or depth-independent boundary;
7. run each claim through the untouched, sealed admission engine;
8. independently reconstruct the result and execute adverse controls;
9. preregister, seal and compare against external observations wherever the claim has an observable consequence; and
10. publish a standalone foundational branch paper only after the foundational inventory and paper evidence map pass their gates.

The foundational paper must say exactly what is established and exactly what is not yet covered. It must include the full derivation chain, candidate census, rejected alternatives, controls, empirical results, limitations and the field-completion roadmap. It is not permitted to describe the whole field as complete merely because its foundational layer is complete.

Layer Two - complete field reconstruction

After the foundational paper, the branch expands across the complete registered field census. Subject families are developed in dependency order and in coherent batches, but every claim receives its own package, admission decision, receipt and evidence. A branch reaches current-knowledge closure only when:

- every registered current-knowledge obligation has exactly one owner;
- every branch-owned prior result has a same-strength V3 disposition;
- every derivable law has a root trace to the foundational theorem;
- every assembled law is forced across its declared complete grammar;
- every measurable consequence possible at the declared boundary is tested through the preregistered post-seal empirical route;
- favourable, adverse, absent and unresolved results are all preserved;
- the complete live census equals the paper evidence map; and
- the branch publication gate passes without altering any protected authority.

The resulting paper is a new version of the same branch paper, not a replacement record that hides earlier boundaries. Major subject-family completions may produce intermediate versions. The current-knowledge-complete version remains open to lawful extension, correction, falsification and later discovery.

Building a complete field census

"Cover the field" cannot mean an informal list assembled while derivations are already under way. Before Layer Two begins, each branch must freeze a dated, source-bound field census built from:

1. every V1/V2 theorem, value, prediction, caveat and named open question;

2. the complete terminology and subject hierarchy of authoritative professional or public scientific reference bodies available at the freeze date;
3. standard research and teaching taxonomies, decomposed into scientific questions rather than imported answers;
4. current measurement catalogues, reference datasets and known anomaly lists;
5. explicit cross-branch handoffs and duplicate-owner resolution;
6. historically marginalized, disputed or "fringe" questions stated neutrally with their exact possible evidence, rather than silently omitted or admitted;
7. negative results, null searches, unavailable measurements and standing predictions; and
8. a reviewer extension route for missing obligations discovered after freeze.

Every source entry is reviewed even when the branch owns none of it. Every owned compound statement is decomposed into atomic obligations. The frozen census receives a hash and a dated boundary before it can be used to claim coverage. New obligations enter through a later version; they do not rewrite the old boundary.

Physics-level depth parity

"The same strength and depth as Physics" means that every branch must provide, at minimum:

- a complete categorical-owner audit;
- a dependency-ordered machine census;
- cold-readable WHY, DERIVATION and CHECK sections for every law;
- complete generated alternatives and one-for-one decisions;
- unique-survivor, minimality and exact closure-boundary proofs;
- root traces to *There is no nothing*;
- implementation-distinct reconstruction;
- premise, candidate, artifact and measurement tamper controls;
- preregistered post-seal comparison against every practicable external measurement consequence;
- explicit preservation of agreements, mismatches, nulls and unavailable tests;
- family integration certificates and a final branch-wide structural and empirical reconciliation certificate;
- a manuscript evidence map equal to the live branch inventory; and
- headline presentation of exact scientific results and measured comparisons, with hashes and filenames serving as supporting evidence rather than the headline itself.

Rules shared by every branch

- The scientific dependency spine begins with *There is no nothing*, the One, exact positive finite count and part, and the Fold.
- No numerical null object, negative magnitude, irrational or imaginary proof scalar, completed infinity, ungenerated continuum, fitted parameter or answer-producing consensus model enters derivation.
- The glyph `0` may represent structural absence at a user or host boundary; it does not create a numerical zero in the SFT proof domain.
- Conventional terminology defines questions and correspondence boundaries; it does not select the SFT law.
- Earlier SFT work is a mandatory observational and reconciliation corpus, not executable proof authority for V3.
- Observation is part of empirical science. When an external comparison is possible, the source identity and target are registered before release, the prediction is sealed before target access, and all results are retained.
- Applications and engineering may test admitted laws but cannot choose them.
- A halt remains a halt. Neither the engine nor a verifier may be altered to obtain acceptance.

- Publication and remote mutation require Maria Smith's explicit, action- specific authorisation.

Dependency and handoff order

The spine and its major downstream forks are:

1. Foundation
2. Mathematics
3. Information Science
4. Classical Computational Science
5. Reversible and Quantum Computation
6. Physics
7. Chemistry
8. Materials Science
9. Biology and Life Sciences
10. Medicine and Health Sciences
11. Consciousness and Cognitive Science
12. Earth and Environmental Sciences
13. Astronomy and Cosmology
14. Social and Collective Systems
15. Engineering Translation

This list is an execution order for shared foundations, not a claim that every later science has only one immediate parent. Astronomy consumes Physics and later Chemistry; Earth science consumes Physics, Chemistry, Materials and Biology; Medicine consumes Biology, Chemistry, Materials, Information and Computation; consciousness consumes Biology, Information and Computation. Every cross-branch dependency must identify the admitted upstream receipt and must not rederive or relocate the upstream law.

Paper lifecycle

For every branch:

1. **Roadmap and census edition** - scope, ownership, prior-work audit and empirical source plan; not a scientific paper and not closure.
2. **Foundational paper edition** - published only after Layer One passes.
3. **Subject-family updates** - sequential versions adding admitted families, updated evidence maps and explicit remaining work.
4. **Current-knowledge-complete edition** - issued after the full frozen field census passes, while explicitly inviting lawful extensions.
5. **Later extensions** - new versioned claims and paper updates; old receipts remain immutable.

Version numbering continues from the latest valid record for that branch. It is never reset merely because the roadmap is reorganized.

Branch plans

- [Foundation](00-foundation.md)
- [Mathematics](01-mathematics.md)
- [Information Science](02-information-science.md)
- [Classical Computational Science](03-classical-computation.md)

- [Reversible and Quantum Computation](04-quantum-computation.md)
- [Physics](05-physics.md)
- [Chemistry](06-chemistry.md)
- [Materials Science](07-materials.md)
- [Biology and Life Sciences](08-biology.md)
- [Medicine and Health Sciences](09-medicine.md)
- [Consciousness and Cognitive Science](10-consciousness-cognitive-science.md)
- [Earth and Environmental Sciences](11-earth-environment.md)
- [Astronomy and Cosmology](12-astronomy-cosmology.md)
- [Social and Collective Systems](13-social-collective-systems.md)
- [Engineering Translation](14-engineering-translation.md)

Status language

Roadmap checkboxes are prohibited as evidence of scientific completion. Each branch plan names work and gates only. Live counts, last admitted claims and next operations belong in the branch continuation checkpoint and machine census so a conversation restart cannot turn an old plan into a false status report.

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Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 0.3.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

This paper introduces the third clean-room reconstruction of Smithian Fold Theory (SFT) and the Ernos Labs open-science platform built to make each SFT claim independently inspectable, executable, reproducible and falsifiable. The project begins

from a premise-free operational theorem: **there is no nothing**. At the declared proof boundary, a purported counterexample either presents no statement, denial, record or proof object and therefore supplies no counterexample, or it presents one and that presentation is an occurrence rather than nothing. A complete two-class partition closes this operational claim without an axiom, fitted parameter, numerical zero, negative quantity, irrational value, imported theory or earlier SFT answer. From the admitted occurrence, a complete six-form coverage-and-addition census forces the structural One as the unique complete representation with neither omission nor unforced addition. This result is structural only. Arithmetic, parts and Fold operations were not premises of the founding run; their later admissions remain separate branch evidence.

The paper also specifies a fail-closed admission engine intended to govern the future reconstruction. Every claim must register its dependency chain, provenance class, generated candidate grammar, exact boundary, completeness certificate, decision for every candidate, unique survivor, minimality, named-shape uniqueness, adverse controls and a distinct implementation check. Natural-science claims add preregistration, target-inaccessible prediction, pre-comparison sealing, separate target custody, post-seal measurement, falsification criteria and preservation of unfavorable results. One public command validates repository policy, unit and end-to-end behavior, 100% of the core engine's executable lines, every registered derivation, controls, independent validators, receipts and the ordered census. The percentage is a precise software-coverage statement, not a claim that tests eliminate all possible defects.

The platform is a response to documented structural weaknesses in scientific communication and selection: paywall and author-fee inequality, publication bias against null results, sponsor-associated outcome and agenda effects, variable reliability in grant review, and the limited scientific interpretability of black-box prediction. It does not replace domain expertise, peer criticism, measurement or conventional correspondence. It removes credentials, consensus and predictive opacity from the *admission rule* for an SFT law. Code is released under Apache 2.0; papers and documentation are released under CC BY 4.0. Maria Smith retains copyright. The Ernos Labs name is a conformance designation, not a restriction on lawful copying, forking, testing or criticism.

Keywords: Smithian Fold Theory; open science; computational proof; premise-free foundation; exact arithmetic; reproducibility; falsifiability; clean-room replication; scientific software; knowledge tree.

Scope and ownership boundary

This paper owns its methodological or comparative argument and no downstream scientific branch. Ownership identifies responsibility for derivation and falsification; it does not prevent criticism or lawful cross-branch use. Application performance, consensus, credentials and Lean acceptance cannot transfer ownership or select a law.

Mathematical, computational and empirical constitutions

The mathematical constitution retains exact generated carriers, relations, candidate grammars, decisions, survivors and closure boundaries. The computational constitution retains executable identities, complete traces, controls, independent reconstructions and receipts. The empirical constitution retains source registration, transport, pre-seal and post-seal chronology, custody, measurement or observation, uncertainty, favourable and adverse rows, missingness, unresolved records and falsification conditions. None of these evidence classes substitutes for another.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Data and code availability

The successor Markdown is `publications/successors/methods/THERE_IS_NO_NOTHING_METHODS_PAPER_001_V0_4.md` and its scientific predecessor is `publications/successors/methods/THERE_IS_NO_NOTHING_METHODS_PAPER_001_V0_3.md`. The Lean sources and report are under `generated/lean4_validation/`; the report is `generated/lean4_validation/reports/whole_model_validation.json`. Current claims are indexed by `census/claims.json`; complete candidates, decisions, controls, certificates, observations and receipts remain under `claims/<claim-id>/` and `receipts/`. The paper's evidence map and draft deposit metadata sit beside the successor source. No remote action was performed.

Successor machine-identity appendix

- Preceding source: `publications/successors/methods/THERE_IS_NO_NOTHING_METHODS_PAPER_001_V0_3.md`
- Preceding source SHA-256: `sha256:d427d9803b04001ec6b83fd0612ea7ee7c894d172d8970bd3dc46c17bb4ea19f`
- Lean report: `generated/lean4_validation/reports/whole_model_validation.json`
- Lean report SHA-256: `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`
- Ordered census SHA-256: `sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71`
- Execution manifest SHA-256: `sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41`
- Protected engine seal: `sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a`
- Verification-authority seal: `sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8`
- Publication authorised: `false`
- Remote actions performed: `false`

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned `PASS`. Its immutable report identity for this successor is `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper is a methodological or comparison paper and does not acquire ownership of downstream claims through this verification.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar: `unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter. Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors. `#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted gate` only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 0.4.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes methodological or comparative analysis and does not acquire ownership of a scientific branch.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.